A Two-Stage Learning Approach for Solar Flare Classification and Time-Series X-Ray Flux Prediction

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***Abstract*—Solar flare prediction has been commonly framed as a classification problem, with algorithms focused on forecasting whether an active region will or will not produce a flare within a future time window. However, the process of a flare and its evolution, the component most directly connected to operational risk, cannot be deduced solely from binary classification. This paper presents a two-stage framework that integrates magnetic feature-based flare classification with time-series X-ray flux forecasting. The first stage trains an XGBoost classifier on aggregated SHARP magnetic parameters and GOES soft X-ray flux features to predict non-flaring, M-class, and X-class activity. The second stage applies sequence-to-sequence LSTM regression to forecast future GOES 1-8 Angstrom X-ray flux trajectories for samples classified as flaring. Evaluation on the SWAN-SF dataset demonstrates approximately 98 percent classification accuracy in Phase 1 and approximately 16 percent peak-magnitude forecasting error in Phase 2, with inference times below 0.6 seconds per sample. By combining occurrence prediction with flux evolution forecasting, this improved framework provides a more informative and deployable approach for real-time space-weather forecasting.**

***Index Terms*—Machine learning, solar flare prediction, time-series forecasting, sequence-to-sequence learning, XGBoost classification**

I. Introduction

Solar flares are rapid releases of magnetic energy from the Sun that can disrupt satellites, aviation communications, navigation systems, and ground-based power infrastructure [12]. Flares are classified by their peak X-ray brightness, where M- and X-class events represent the most intense categories and pose the greatest concern for space-weather monitoring [3].

Most existing machine learning approaches to solar flare forecasting rely heavily on binary or multi-class classification. Under this methodology, magnetic or radiative observations from an active region are used to predict whether a flare of certain intensity will occur within a specified future window. And while this approach is valuable for event detection, it does not directly model the continuous evolution of soft X-ray flux, which is one of the clearest observational measures of flare intensity and severity. On the other hand, methods that focus only on X-ray flux regression often fail to incorporate magnetic and physical drivers that are closely associated with flare initiation. These differences leave an important research gap between predicting flare occurrence and characterizing the potential severity of the event.

To address this gap, the following study introduces a two-stage physics-guided machine learning framework that combines multiclass flare classification with future X-ray flux forecasting. The first stage uses XGBoost on engineered statistics from SHARP magnetic parameters and GOES soft X-ray measurements to provide accurate, interpretable classification, while the second uses a sequence-to-sequence LSTM to predict the temporal evolution of flare intensity under near-real-time inference conditions. The dual-phase framework is then evaluated on the basis of classification performance, feature ablation, SHAP-based interpretation, regression error, baseline comparison using Taylor diagrams, and inference latency.

II. Related Work

Early flare forecasting approaches relied on statistical or physics-inspired methods based on historical data, those of which included Poisson flaring probabilities, McIntosh sunspot classifications, and active-region monitoring [4]. These methods provided useful probabilistic estimates, but they often struggled to generalize to complex and rapidly changing magnetic field configurations. With the advent of computing, modern flare forecasting has increasingly adopted machine learning techniques due to solar observations naturally producing high-dimensional, multivariate, and temporally structured data.

The forecasting approach introduced in this study builds on the advantages of structured-data classification while extending the model beyond flare detection alone. By adding a second stage that predicts the future X-ray flux trajectory, the framework connects the flare occurrence likelihoods with the flare's expected radiative evolution. This design is especially suited for scientific machine learning applications, where models not only achieve strong predictive performance, but also remain interpretable for practical deployment.

III. Data Acquisition and Processing

This study uses the SWAN-SF dataset, which contains multivariate time-series records of solar active regions observed between May 2010 and December 2018 [1]. Each record includes Spaceweather HMI Active Region Patch (SHARP) quantities derived from Solar Dynamics Observatory vector

magnetograms, which are then combined with GOES soft X-ray flux measurements. SHARP parameters provide magnetic and geometric information about active regions, including magnetic field components, current helicity, Lorentz-force-related quantities, and quality indicators, while GOES 1-8 Angstrom X-ray flux provides the radiative signal used to define flare magnitude [8].

Since the forecasting algorithm is separated into two distinct phases, the time-series datasets are processed and aggregated according to the requirements of each phase. In Phase 1, the classification model receives a flattened window of historical flare activity from a given solar active region and uses it to categorize the solar region's behavior into one of three class labels: non-flaring, M-class, or X-class. In Phase 2, the regression model receives the full sequential form of the historical time-series window and forecasts the incoming future sequence of X-ray flux evolution.

Input windows were set to $L = 15$, where each time step represents 12 minutes, resulting in three hours of historical observations. The classification horizon is set to $H = 15$, corresponding to a three-hour future window, while the regression horizon is set to $H = 5$, corresponding to one hour of future soft X-ray flux. Fig. 1 shows an example full active-region X-ray flux life cycle, and Fig. 2 shows the extracted input and output windows used for the flux forecasting task.

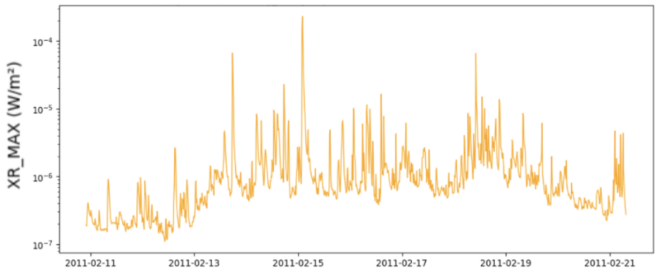


Fig. 1. Full active-region GOES X-ray flux life cycle before extracting fixed-length model windows.

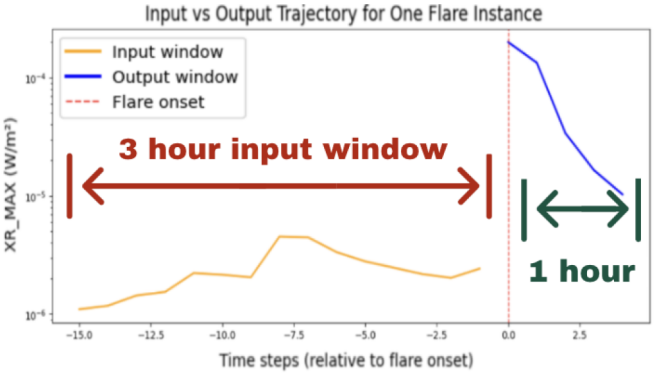


Fig. 2. Extracted three-hour input window and one-hour forecast horizon used for the flux forecasting stage.

Flaring samples are collected from intervals containing M-class or X-class activity. To increase coverage of flare-associated behavior, positive samples are obtained through both flare-centered sampling and random sampling within flare-associated intervals. Non-flaring samples are drawn from periods without M-class or X-class events and are used exclusively for Phase 1 classification. After filtering and sampling, the dataset contains 7,976 entries: 4,913 non-flaring samples, 2,882 M-class samples, and 181 X-class samples.

Upon allocating samples into their classes, preprocessing methodologies were implemented, those of which included removing missing values, thresholding quality indicators, and applying logarithmic scaling to magnetic, geometric, and radiative features. Samples with unreliable magnetic field information or faulty X-ray measurements were filtered to reduce projection effects and preserve data integrity. Given the range of quantitative values, logarithmic scaling was leveraged to account for both the magnetic quantities and X-ray flux features spanning multiple orders of magnitude.

For model training, the processed datasets were randomly divided into training, validation, and testing sets. Although an active-region-based split would better reduce dependence on region-specific patterns, subsequent feature-importance analysis was conducted after model evaluation to assess whether geographic or region-identifying variables substantially influenced the predictions.

IV. Methodology

A workflow diagram in Fig. 3 illustrates the proposed architecture. The structure presented optimizes computational efficiency because Phase 2 regression is not applied to every non-flaring sample; concurrently, the system improves interpretability because occurrence prediction is separated from intensity evolution.

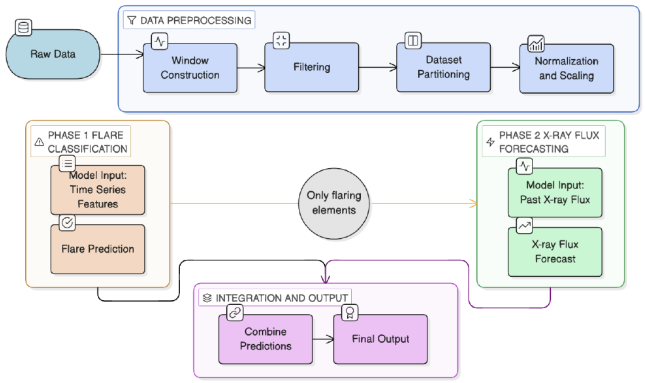


Fig. 3. Workflow of the two-stage solar flare classification and X-ray flux forecasting framework.

All model training was conducted using Kaggle's publicly accessible, free TPU v3-8 environment, resulting in no direct computational cost. Hyperparameters were tuned using the training and validation sets, while the testing sets were reserved for model evaluation.

A. Fourier Analysis

Before model development, Fourier analysis was applied to the curated GOES X-ray flux time-series dataset to investigate whether flare behavior can be characterized by stable frequency components [9]. Upon decomposing the flux signals via Fourier transforms, the resulting spectrogram in Fig. 4 reveals how flare events appear as broadband power enhancements rather than narrowband oscillations, while Fig. 5 illustrates how simple frequency filtering can distort the flare signal. As a result, the Fourier results suggest that flare prediction is unlikely to be achieved through narrowband frequency detection and instead requires analysis of nonlinear temporal evolution or magnetic field properties, solidifying the necessity of physical and computational modeling.

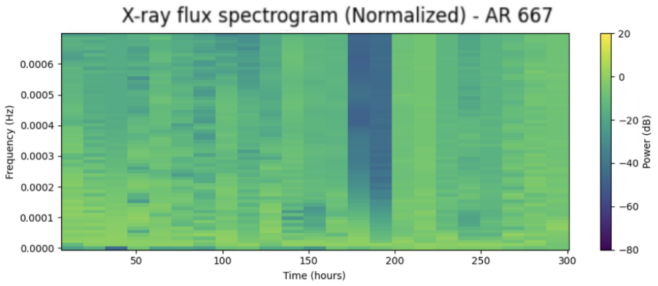


Fig. 4. Fourier spectrogram of GOES X-ray flux showing broadband power enhancement during flare activity.

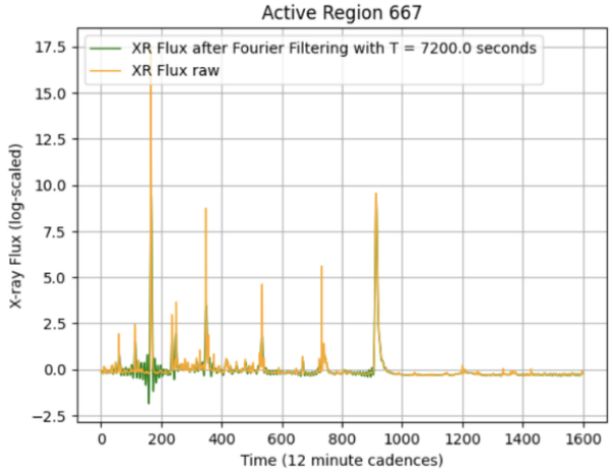


Fig. 5. Comparison of raw GOES X-ray flux and the signal obtained after suppressing low-frequency Fourier components for active region 667. The filtering removes components with periods longer than 7200 seconds and suppresses part of the impulsive flare profile. Instead of reducing noise, perturbations became more obvious and volatile, illustrating why fixed frequency filtering is not used as the forecasting model.

B. Phase 1: XGBoost Classification

The first stage leverages Extreme Gradient Boosting (XGBoost) to perform multiclass classification [2]. Due to the fact that XGBoost operates on tabular inputs, each multivariate time series was first transformed into a two dimensional fixed-length feature vector: for each SHARP parameter and X-ray flux feature, the mean, standard deviation, minimum, maximum, and final value are computed over the input window. This aggregation retains information about the level, spread, extremes, and latest state of each physical variable while producing a representation compatible with gradient-boosted trees.

The XGBoost model in this study is configured with 100 decision trees, a maximum depth of 6, a learning rate of 0.1, and a multinomial log-loss objective. XGBoost is an ensemble tree-based method, meaning that it combines many decision trees in sequence, where each new tree is trained to correct the errors made by the previous trees. This makes it well suited for tabular inputs, especially when the features come from a variety of physical quantities such as magnetic field strength, current helicity, Lorentz-force-related parameters, and soft X-ray flux. Because each engineered input feature represents a summary of the active region's recent temporal behavior, XGBoost can learn nonlinear relationships between these variables without the complexity of the full time-series.

Furthermore, a tree-based model is selected for the classification stage because the input representation is both physically meaningful and interpretable. XGBoost can be paired with feature-importance analysis to evaluate whether the classifier is relying on magnetic and radiative variables that are physically connected to flare activity. This is especially important in solar flare forecasting, where strong predictive accuracy alone is not sufficient; the model should also demonstrate that its predictions are informed by physically meaningful properties rather than incidental patterns in the dataset.

In this study's case, after the XGBoost model is trained on the training dataset, feature importance is then evaluated with Shapley Additive Explanations (SHAP) [6]. SHAP values estimate how much each feature contributes to a prediction relative to a baseline expectation. Aggregating SHAP values over the testing set then provides a global ranking of all variables and their measured importance.

C. Phase 2: Seq2Seq LSTM Forecasting

The second stage uses sequence-to-sequence Long Short Term Memory (LSTM) regression [10] to forecast future GOES X-ray flux trajectories. Given flares' different temporal scales and variability, two separate models are trained in this phase, one for M-class flares and one for X-class flares. Fig. 6 illustrates the encoder-decoder structure.

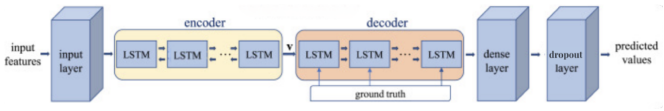


Fig. 6. Seq2Seq encoder-decoder LSTM architecture used for flux forecasting.

The encoder uses the historical flux sequence to form a latent representation of the observed flare behavior, which the decoder then uses to predict future flux values step by step. During training, teacher forcing is applied, where the ground-truth value from the previous step is passed to the decoder

instead of the model's own previous prediction [5]. This helps stabilize learning and improves the model's ability to capture short-term temporal dependencies. During testing, however, the model does not utilize the ground-truth values and operates autoregressively, meaning each predicted value is fed back as the next decoder input. This setup better reflects real-time forecasting, where future ground-truth values are unavailable.

In addition to the encoder-decoder structure, several training strategies are used to improve generalization. Huber loss is applied because it is less sensitive to extreme errors than mean squared error while still penalizing large deviations in the predicted flux trajectory. To reduce overfitting, early stopping monitored the validation loss with a patience of 8 epochs and restored the weights from the epoch with the lowest validation loss. In addition, the learning rate was reduced by a factor of 0.5 when the validation loss failed to improve for 4 consecutive epochs. An attention mechanism was also implemented to improve temporal alignment and learning between the input and output sequences [11]. Ultimately, Phase 2 models were trained for a maximum of 100 epochs but typically stopped at around 70 due to the criteria set by early stopping.

Because intense flaring samples are limited, Gaussian noise augmentation is applied to the encoder inputs to encourage more stable latent representations and expand the training set by a factor of 100 [7]. Fig. 7 summarizes how different Gaussian noise levels affect regression behavior.

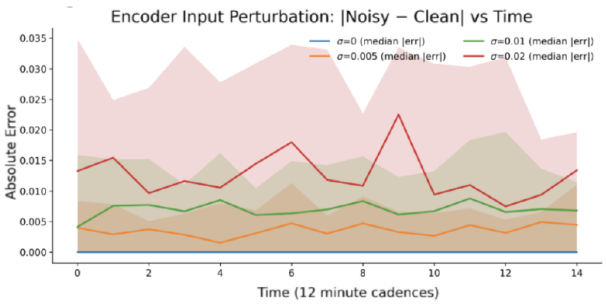


Fig. 7. Gaussian noise augmentation analysis for encoder inputs.

V. Experimental Results

All results are evaluated on unseen testing data that was excluded from both the training and validation stages. This separation ensures that the reported performance reflects the model's ability to generalize to new samples rather than memorize patterns from the training set. The evaluation includes several components: classification performance, feature interpretation, feature ablation, time-series forecasting accuracy, baseline comparison, and inference efficiency. Together, these metrics assess not only whether the framework can identify flare classes accurately, but also whether its predictions are interpretable, stable under reduced feature sets, effective for flux forecasting, and practical for real-time deployment.

Direct comparisons to existent research remain limited, as prior studies do not appear to present a unified two phase pipeline like the one described in this paper. As a result, the discussion in this section is intended to establish methodological comparisons rather than suggest a direct numerical improvement over previous flare-forecasting methods.

A. Classification Performance

The Phase 1 XGBoost classifier achieves approximately 98 percent overall accuracy on unseen testing data, demonstrating strong performance in distinguishing non-flaring, M-class, and X-class samples. As shown in Fig. 8, the model correctly classifies approximately 79 percent of the rare X-class flare samples, 98 percent of the M-class samples, and 98 percent of the non-flaring samples. The disparities between the performance of non-flaring samples and X-class flares is especially relevant because X-class flares are substantially underrepresented in the dataset, making them more difficult to classify reliably. These results suggest that the aggregated magnetic and radiative features provide a strong and informative representation of active-region behavior, allowing the classifier to maintain acceptable accuracy even under class imbalance.

TABLE I
Classification performance of the XGBoost model.

Class	Precision	Recall	F1-score
0	0.99	0.98	0.99
1	0.96	0.98	0.97
2	0.90	0.79	0.84
Macro Avg.	0.95	0.92	0.93
Weighted Avg.	0.98	0.98	0.98

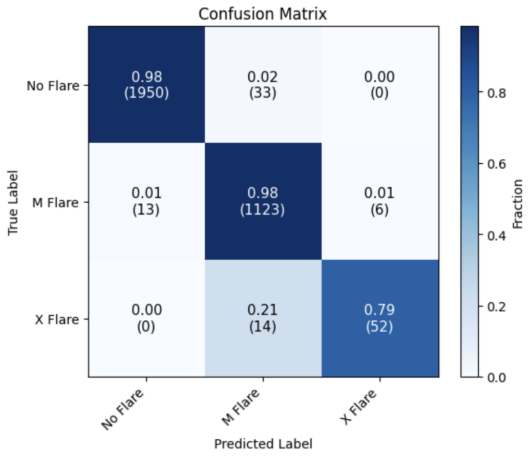


Fig. 8. Confusion matrix for the XGBoost multiclass flare classifier.

Feature ablation analysis further evaluates whether the classifier depends on many weak features or a smaller set of highly informative variables. ROC comparisons across feature subsets show that the model maintains strong separability even when the number of input features is reduced, as shown in Fig. 9, Fig. 10, and Fig. 11. These results indicate that the XGBoost model aligns with the physics-backed principle that magnetic and radiative predictors capture much of flare behavior.

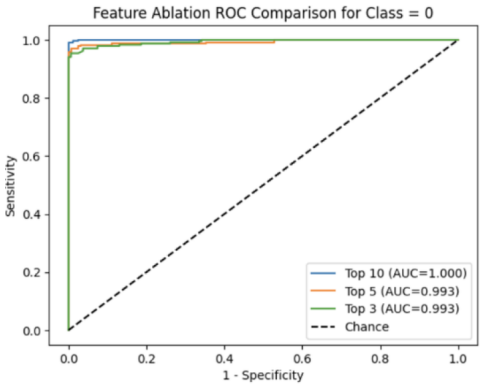


Fig. 9. ROC comparisons for the non-flare class.

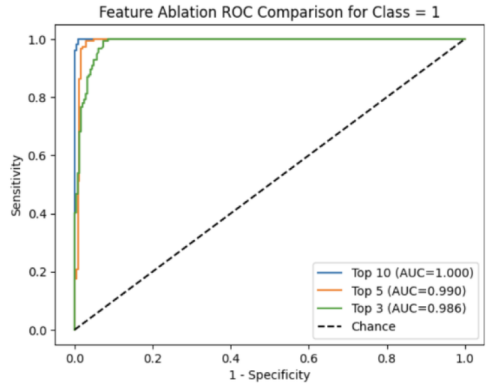


Fig. 10. ROC comparisons for the M-flare class.

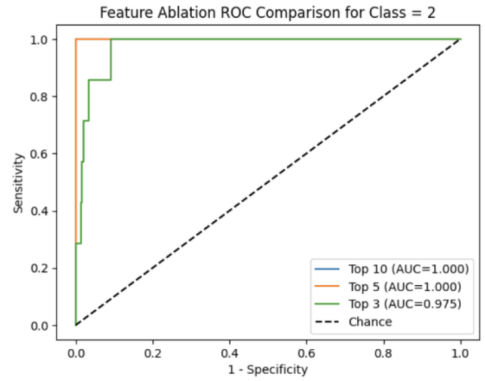


Fig. 11. ROC comparisons for the X-flare class.

B. SHAP-Based Feature Interpretation

The global SHAP feature importance plot in Fig. 12 shows that the most influential predictors include total unsigned magnetic flux, Lorentz-force-related SHARP quantities, and current helicity. These variables are physically meaningful because they are associated with magnetic non-potentiality, free energy accumulation, and stress within active regions. The absence of simple region-identifying features among the dominant predictors reduces the likelihood that the classifier is merely memorizing active-region identity.

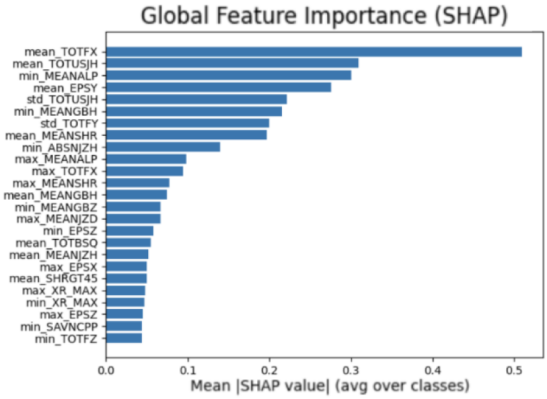


Fig. 12. Global feature importance ranked by SHAP values for the classification stage.

Class-specific SHAP summaries, shown in Figs. 13–15, provide additional evidence that feature importance varies by flare class. Therefore, the multiclass classifier relies on class-specific indicators rather than identical feature patterns. Holistically, this SHAP analysis provides both interpretability and a physics consistency check for the classifier.

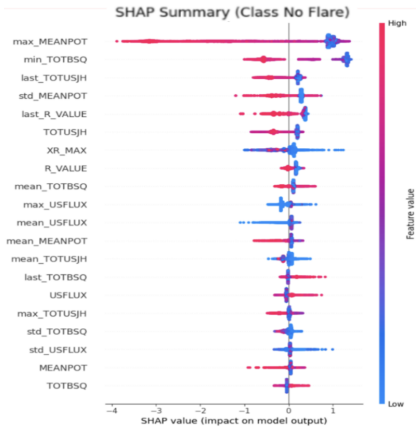


Fig. 13. Class-specific SHAP summary for the non-flaring class.

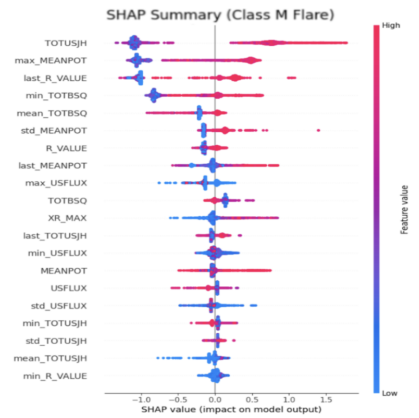


Fig. 14. Class-specific SHAP summary for the M-class flare class.

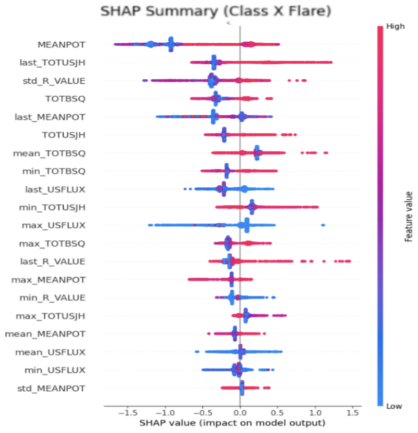


Fig. 15. Class-specific SHAP summary for the X-class flare class.

C. Forecasting Performance

The Phase 2 Seq2Seq LSTMs were able to successfully reconstruct the temporal evolution of soft X-ray flux for both M-class and X-class flare samples. Aggregate forecasts on unseen testing data are shown in Fig. 16. In the graphs, the predicted mean flux closely follows the true mean flux across the forecast horizon, demonstrating the seq2seq model's heightened temporal forecasting abilities.

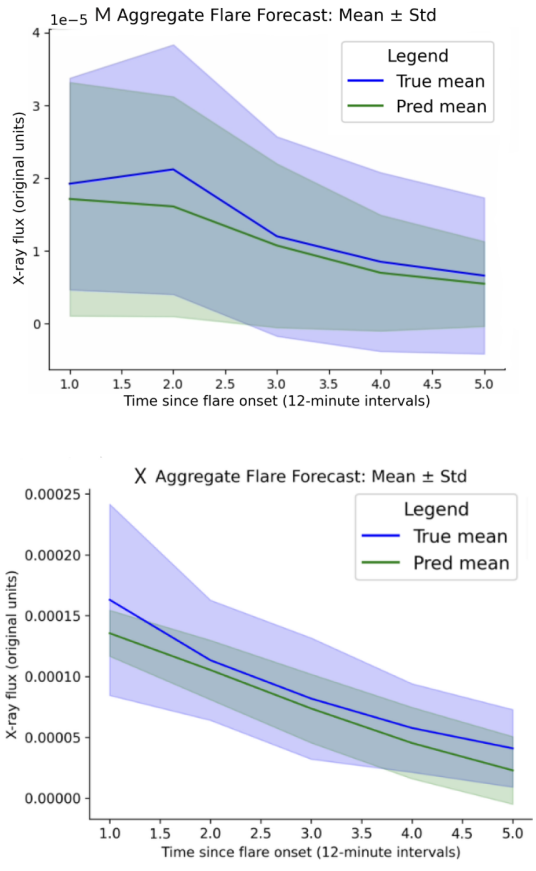


Fig. 16. Aggregate forecast on unseen testing data for the Seq2Seq LSTM inference model.

Quantitatively, the model achieves an RMSE of 5.56×10^{-5} for X-class forecasts and 1.40×10^{-5} for M-class forecasts, with these values corresponding to an error of approximately 16% of peak flux magnitude. Log-space RMSE values of 0.44 for X-class flares and 0.38 for M-class flares also indicate stable predictive behavior across multiple orders of magnitude in X-ray flux intensity. The slightly weaker X-class trajectory match and performance metrics are consistent with the scarcity and higher variability of extreme flares.

D. Baseline Comparison

To determine the relative effectiveness of the Seq2Seq LSTM, the model is compared against both a persistence baseline, where the baseline repeats the last observed flux value across the forecast horizon, and a linear regression model, where the output flux is linearly fit. Taylor diagrams are leveraged to compare correlation, standard deviation, and RMSE in a single visualization, as shown in Fig. 17. As shown by its smaller radial distance, the study's LSTM model lies closer to the observation reference point than the persistence and linear regression baselines for both M-class and X-class forecasts, indicating higher correlation and lower variance error.

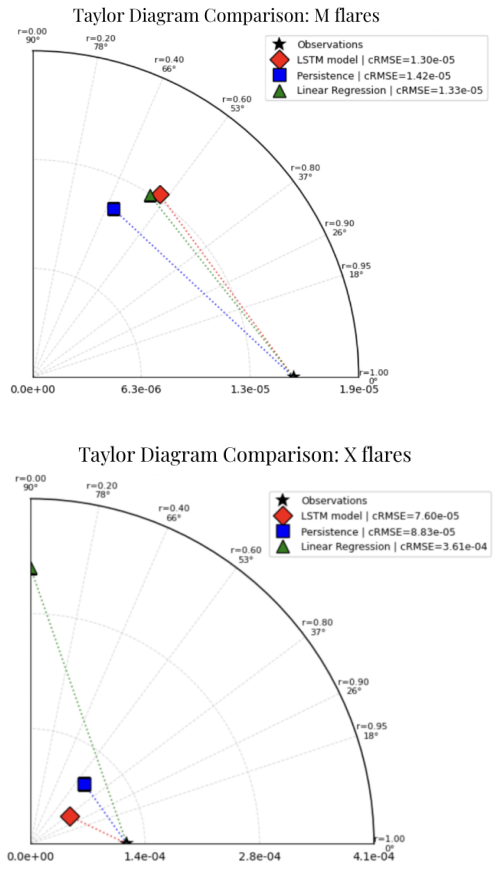


Fig. 17. Taylor diagrams comparing Seq2Seq LSTM forecasts with persistence and linear-regression baselines.

E. Efficiency and Deployment Relevance

The study's two phase flare forecasting algorithm is designed for low-latency inference. From the performance metrics collected during training and testing, the autoregressive forecasting stage runs in less than 0.6 seconds per sample, with approximate inference times of 0.49 seconds for X-class forecasts and 0.59 seconds for M-class forecasts. Because the regression model is only executed after the classification stage identifies a likely flare, the pipeline avoids unnecessary sequence prediction on non-flaring samples. This improves scalability and supports operational use cases where many active regions must be monitored continuously.

VI. Discussion

The proposed framework strengthens solar flare forecasting by combining structured-data classification with sequence modeling. From a machine learning perspective, the approach uses XGBoost for interpretable classification and Seq2Seq LSTM regression for temporal forecasting. From a physics perspective, the dominant features identified by SHAP align with magnetic energy buildup and stress, whereas the regression stage models the short-term radiative evolution of flares.

A key advantage of the two-stage design is that it produces richer information than classification alone. Classification determines whether a flare is likely, while flux forecasting estimates how the flare intensity may change. This methodology aligns more closely with operational needs, as mitigation decisions depend on occurrence probability and expected severity.

The approach also demonstrates why model interpretability matters for scientific machine learning. The SHAP results provide a bridge between predictive features and physically meaningful active-region properties. Rather than treating the classifier as a purely statistical black box, the feature rankings help verify that the model is using quantities connected to flare-producing magnetic stress.

The system still comes with limitations, however. X-class flares are rare, making extreme-event forecasting difficult. The framework uses physics-guided inputs but does not explicitly enforce physical constraints in the loss function. In addition, the current random data split may allow active-region-specific patterns to appear across training and testing sets; future work should therefore evaluate active-region-based partitioning to reduce the risk of information leakage. Broader comparisons with state-of-the-art flare-classification and time-series forecasting models would also help establish the relative performance of the framework. Further extensions may include full vector magnetogram images, hybrid CNN-LSTM architectures, physics-informed neural networks, and magnetohydrodynamic constraints to improve predictive robustness and physical consistency.

VII. Conclusion

Overall, this paper presents a two-stage machine learning framework for solar flare classification and X-ray flux forecasting. The first stage leverages XGBoost to classify non-flaring, M-class, and X-class samples from aggregated magnetic and radiative features, while the second applies a Seq2Seq LSTM model to forecast future soft X-ray flux trajectories for flaring samples. Experiments demonstrate approximately 98 percent classification accuracy, improved performance over persistence and linear regression baselines, and sub-second inference time. By connecting flare occurrence prediction with intensity evolution forecasting, this new framework provides a more actionable, efficient, and deployable approach for real-time space-weather monitoring.

Acknowledgment

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